

MANGU 2025 KCSE PREDICTION



CONFIDENTIAL EXAM

231/1

BIOLOGY

PAPER 1(THEORY)

TIME: 2 HOURS



NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

(Kenya Certificate of Secondary Education)

INSTRUCTIONS TO CANDIDATES

- Write your name and admission number in the spaces provided.
- Answer **all** questions in the spaces provided
- Candidates should check the question paper to ascertain all the pages are printed as indicated and no questions are missing.

For Examiners Use Only

1	2	3	4	5	6	7	8	9	10

11	12	13	14	15	16	17	18	19	20

21	22	23	24	25	26	GRAND TOTAL

1. a) Name the **one** chemicals of a life that makes up **protoplasm**
(1mark)

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b) State **Two** disadvantages of using fats as respiratory substrates
(2marks)

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2. (a)Name the type of gene mutation that results in loss of skin **pigment** (1mark)

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b) Explain how sunlight contributes to stronger bones and teeth in human beings (2marks)

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3. Name the main target organ of the following hormones:
(2marks)

a) Antidiuretic.....

b) Glucagon.....

4. a) What is Binary Fission?
(1mark)

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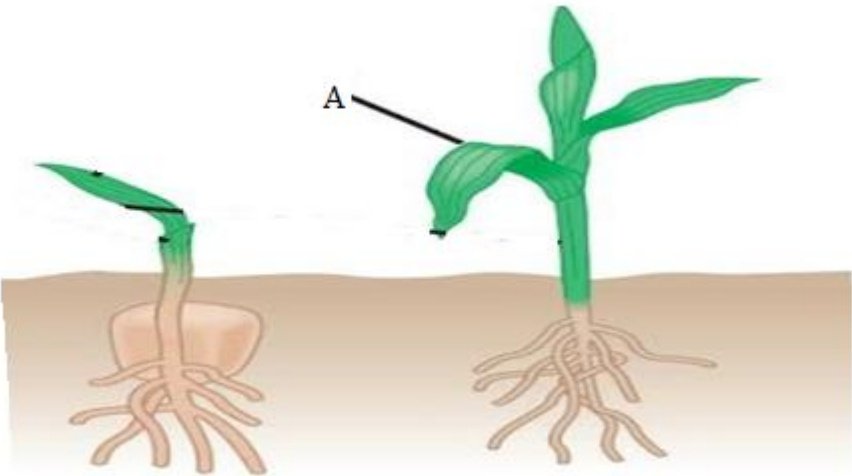
b) Give **Two** disadvantages of sexual **reproduction**
(2marks)

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5. The diagram shown below represents a seedling. Use it to answer questions that follow



a)(i) Name Class in which the seedling **belong**
(1mark)

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(ii) Give reasons for your answer in a (i) above
(2marks)

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b) Explain why the biomass of part labelled **A** will be lower compared to the one found in the seed stage of the same plant

(2marks)

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6. A solution of sugarcane was boiled with hydrochloric acid. Sodium hydrogen carbonate was added and the mixture heated with Benedict's solution. An orange precipitate was formed.

a) Why was the solution boiled with hydrochloric acid?

(1mark)

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b) To which class of carbohydrates does sugarcane belong?

(1mark)

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.....

c) Name the type of reaction that takes place when:

Simple sugars combine to form complex sugar.

(1mark)

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d) State the form in which carbohydrates are stored in **animals**

(1mark)

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7. a) State **Two** ways in which blood clotting is important to a human being

(2marks)

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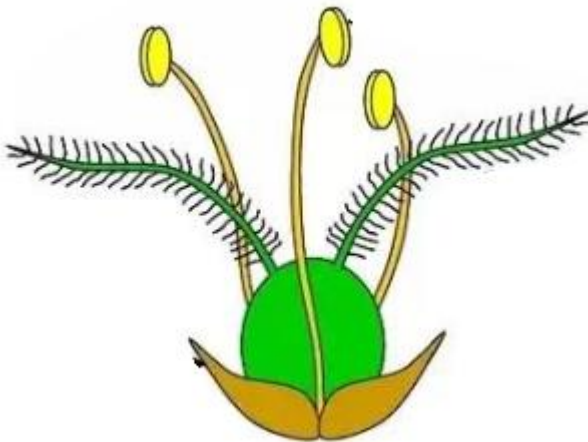
b) What are the roles of thrombokinase enzyme during blood clotting?
(2marks)

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8. The diagram shown below represents a flower



a) Name the agent of pollination for the flower shown above
(1mark)

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b) Give **Two** reasons for your answer in a) above
(2marks)

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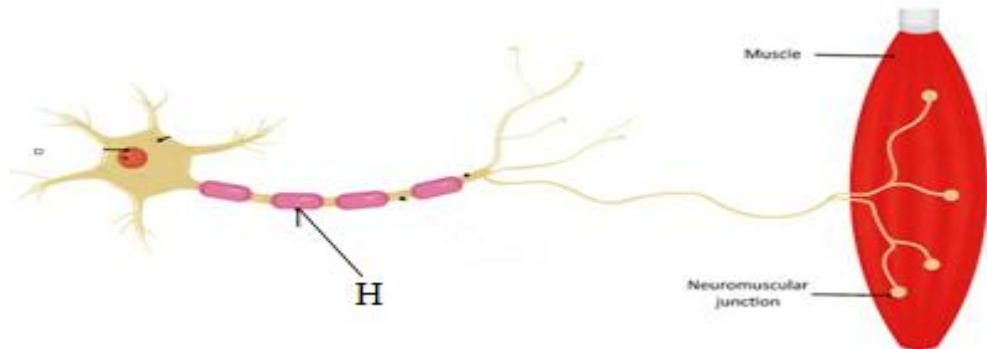
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9. State the differences between cones and rods in terms of the following (2marks)

Feature	Cone	Rod
Visual acuity	High	
Photochemical	Low	

10. Use the diagram of a nerve cell shown below to answer questions that follow



a) With a reason, give the identity of the nerve cell
(2marks)

Identity

Reason

b) Explain the significance of absence of part labelled **H** in nerve cells found in the brain.(1 mrks)

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11. Give **Three** features that make modern man to be more adaptable to the environment
(3marks)

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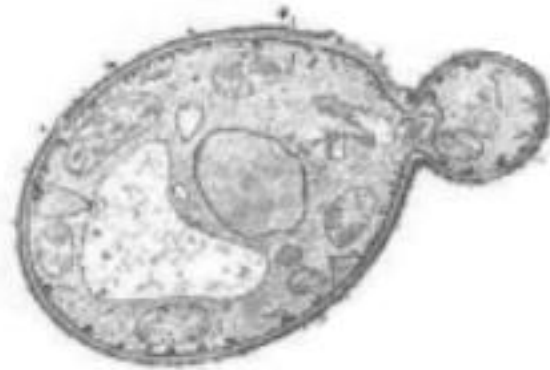
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12. The diagram below represents a living organism



a) State **Two** economic importance of the above organism in the food industry
(2marks)

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b) Why does the rate of respiration reduce under the following conditions?
(2marks)

i) Low temperature

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ii) Metabolic poison

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13. State **Two** reasons why Biotechnology is important in modern science (2marks)

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14. Fill in the table shown below to give differences between biceps and gut muscles **(3marks)**

	Biceps	Gut muscles
I		
II		
III		

15. Explain how variation is important in the process of evolution?

(2marks)

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16. Explain the term negative **feedback**

(1mark)

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17. State **Two** contributions of Carolus Linnaeus (1708 – 1778) to **taxonomy** **(2marks)**

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18. a) In an experiment, Venna counted 9 cells along the diameter of field of view of a light microscope measuring 3.0mm. Determine the diameter of one cell in micrometers **(3marks)**

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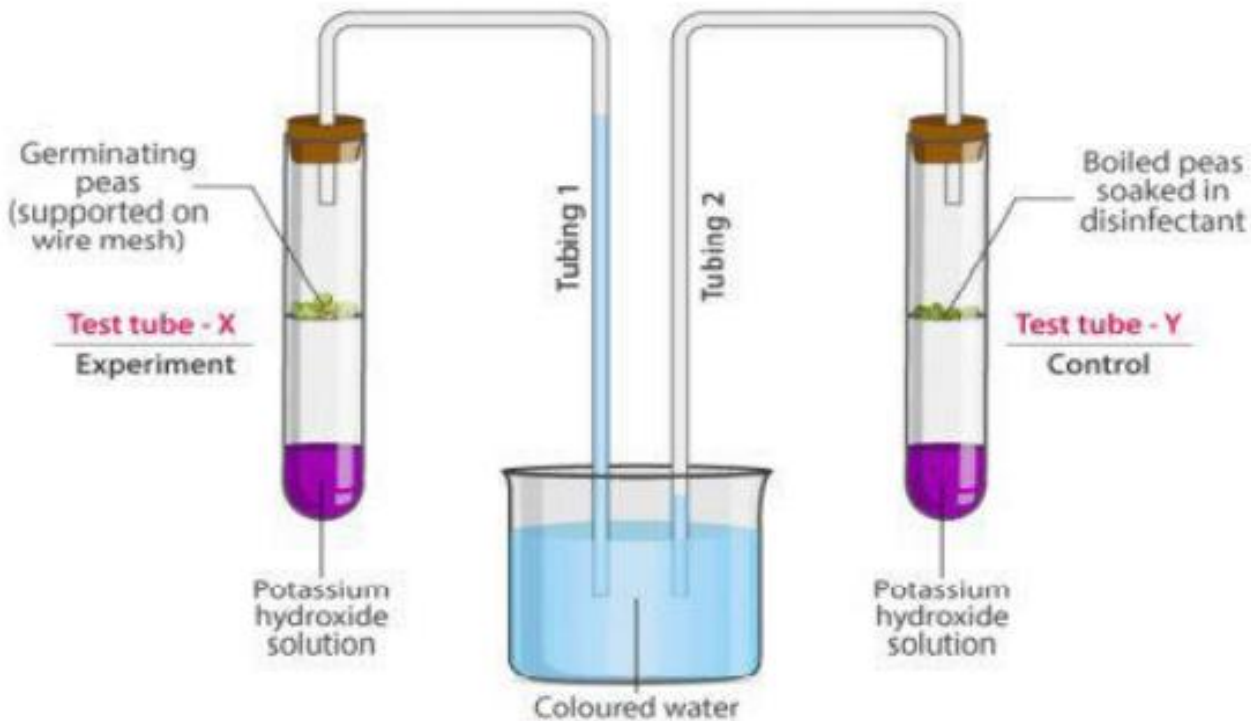
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b) **Why** is electron microscope safer to the eye than light microscope during use? **(1mark)**

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19. The following diagram represents results of an experiment carried out on two sets of germinating seeds.



a) Account for the result shown in test tube X
(2marks)

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b) What is the importance of dipping the boiled seeds in a disinfectant in test tube Y? (1mark)

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20. The diagram shown below represents a section of the vertebral column



a) Name the part labelled S

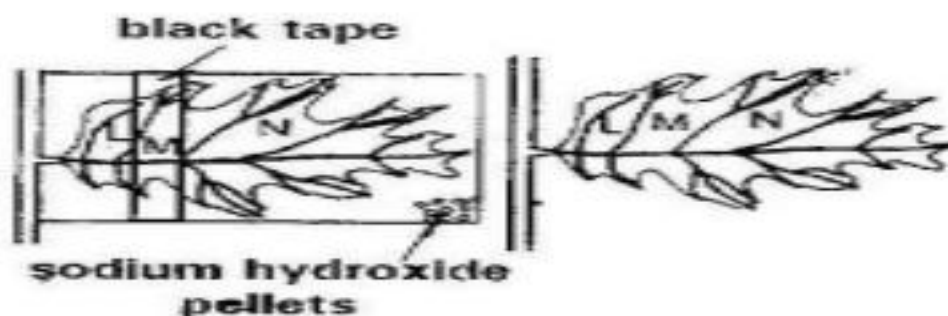
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b) State **TWO** ways in which part T is important to movement in human beings (2 marks)

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21. The diagram below shows an experiment carried out to investigate photosynthesis in a potted plant which has been kept in the dark for 48 hours.



The setup was left in the sunshine for 6 hours. The leaf was tested for starch using iodine solution at the end of the experiment.

a) What would be the colours of the regions of the leaf marked L, M and N?

(3marks)

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b) What is the function of the sodium hydroxide pellets?

(1mark)

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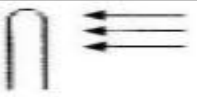
22. (a) In fruit flies (*Drosophila*) the gene for red eyes (R) is dominant over the one for white – eye (r). In a true breeding white – eyed male, all the offspring will be red eyed. However, if a true – breeding white-eyed female is mated with a true- breeding red- eyed male, all the female offspring will be red – eyed. Explain this apparent contradiction. (1mark)

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(b) Work out the expected FI genotypes if a red- eyed female offspring from the cross- described in (a) above is mated with red- eyed males. (4marks)

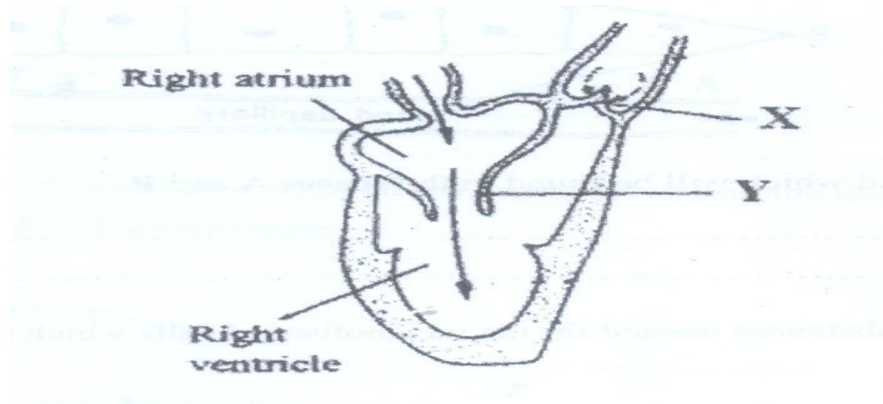
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23. An experiment was set up to investigate the effect of unilateral/unidirectional light in growth of maize seedlings. The diagram in the table below represents experimental set up at the beginning. The tip of D is covered by an aluminium foil.

Experimental set up	Beginning of experiment	Expected results
C		
D		
B		

Using diagrams, complete the table to show the expected results in experimental set up.(3 marks)

24. Use the figure below to answer the questions that follow.



(a)Identify the cardiac cycle represented by the diagram.

(1mark)

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(b)Identify the valves X and Y

(2marks)

X.....

Y.....

25. How does sunken stomata help in lowering transpiration?

(3marks)

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26. a) Name the hormone that stimulate the maturation of the graafian follicles to release a mature ovum in female reproductive cycle. **(1mark)**

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b) Explain why menstruation does not take place after fertilization in human beings. **(2marks)**

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